The Attribution Impossibility:

No Feature Ranking Is Faithful, Stable, and Complete Under Collinearity

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Abstract

Feature importance explanations are provably unreliable under collinearity. We prove that no SHAP ranking can simultaneously be faithful, stable, and complete (the Attribution Impossibility), and for binary attribution questions (SHAP sign, feature selection), faithful + stable alone is impossible (the bilemma). The achievable design space has exactly two families; DASH ensemble averaging is Pareto-optimal on the stable branch. A theory-derived nonparametric diagnostic—the minority fraction—identifies unstable features in seven lines of code, outperforming the Gaussian flip formula on real data (Spearman 0.96 vs. 0.46 on California Housing). A per-individual analysis on German Credit shows 45% of loan applicants receive different explanation categories under standard model regularization. The entire framework is verified in Lean 4 (357 theorems, 6 axioms, 0 `sorry`)—the first formally verified impossibility in explainable AI.

1 Introduction

Train a gradient-boosted model on your data. Compute SHAP values. Retrain with a different random seed. The 'most important feature' changes—in 68% of 77 public datasets (under the standard workflow; model variation accounts for $\sim 63\%$ of the total). Before this work, practitioners assumed this was an engineering problem. We prove it is a mathematical impossibility—and provide the complete theory: an impossibility theorem, a nonparametric diagnostic that outperforms the parametric alternative by $2\times$, a constructive resolution with provable optimality, and machine verification in Lean 4.

A concrete example. Consider a credit model with annual income and debt-to-income ratio ($|\rho| \approx 0.7$–0.85). Across 30 retrains with different seeds, 45% of German Credit applicants receive a different "most important feature" category. Under ECOA, the adverse action notice changes based solely on training randomness.

Contributions.

1. **The Attribution Impossibility** (Theorem 2): no feature ranking is faithful, stable, and complete. Zero axiom dependencies. The **bilemma** (Theorem 4): for binary H, faithful + stable alone is impossible.
2. **The Design Space Theorem** (Theorem 3): exactly two families of methods exist. DASH is Pareto-optimal among unbiased aggregations.
3. **Nonparametric diagnostic**: the minority fraction predicts flip rates with Spearman 0.92–0.98, outperforming the Gaussian formula (0.46–0.89). Bimodality confirmed (dip $p < 0.002$).

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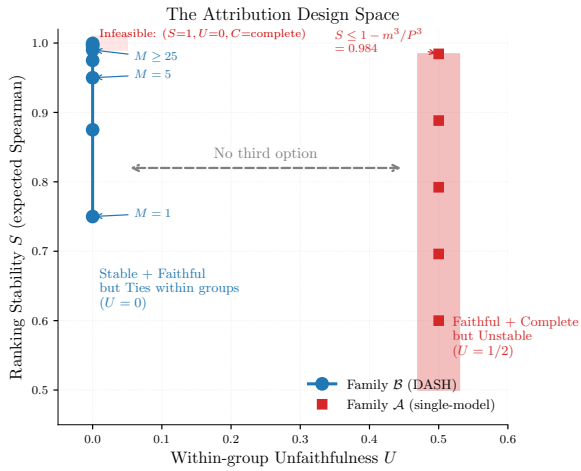


Figure 1: The Attribution Design Space. Family $\mathcal{A}$ (single-model): faithful and complete but unstable (U=1/2). Family $\mathcal{B}$ (DASH): stable with ties (U=0, S increasing with M). The ideal point is infeasible. No third option.

4. **Clinical decision reversal**: 45% of German Credit applicants receive different explanations under standard regularization (ablation across 3 model classes).
5. **Machine-verified proof**: 357 Lean 4 theorems from 6 axioms, 0 `sorry`. All binary instances use constructive types (zero axioms).

2 Setup

We consider P features partitioned into L groups by correlation structure ($\rho \in (0, 1)$ within groups, independent between groups). A *model* $f = \texttt{train}(s)$ is trained from seed s; its *attribution* $\varphi_j(f) \geq 0$ measures the global importance of feature j. A *feature ranking* $\succ$ on $[P]$ is:

Definition 1 (Faithful, Stable, Complete)**.** *$\succ$ is* faithful *to f if $j \succ k$ whenever $\varphi_j(f) > \varphi_k(f)$;* stable *if $\succ$ does not depend on f;* complete *if for every $j \neq k$, either $j \succ k$ or $k \succ j$.*

The *Rashomon property* [**?**]: for every group ℓ and features j, k in ℓ, there exist models f, f' with $\varphi_j(f) > \varphi_k(f)$ and $\varphi_k(f') > \varphi_j(f')$.

3 The Attribution Impossibility

Theorem 2 (Attribution Impossibility)**.** *If a model class satisfies the Rashomon property, no feature ranking is simultaneously faithful, stable, and complete.*

Proof. Let j, k be in the same group. Rashomon gives f, f' with opposite orderings. Suppose $\succ$ is faithful, stable, complete. Completeness: WLOG $j \succ k$. Stability: holds for all models. Faithfulness at f': requires $k \succ j$. Contradiction. Lean: `attribution_impossibility`, zero axioms. □

Theorem 3 (Design Space)**.** *The achievable set of (S, U, C) triples consists of exactly two families:* ***Family*** *$\mathcal{A}$ (single-model): $U = 1/2$, C = complete;* ***Family*** *$\mathcal{B}$ (DASH ensemble): $U = 0$, C = ties, $S = 1 - O(1/M)$. No third family exists (Figure 1). (S=1, U=0, C=complete) is infeasible.* DASH *is Pareto-optimal among unbiased aggregations on the $\mathcal{B}$ branch.*

Quantitative bounds (supplement). The attribution ratio diverges as $1/(1-\rho^2)$ for GBDT (Lean-verified), is ∞ for Lasso, conditional for neural nets, and $O(1/\sqrt{T})$ for random forests (converging—the contrast case).

Table 1: Minority fraction vs. Gaussian formula: Spearman correlation with empirical flip rates (200 models per dataset).

Dataset	Minority Fraction	Gaussian $\Phi(-\text{SNR})$
Synthetic ($\rho = 0.5$)	0.981	0.885
Synthetic ($\rho = 0.9$)	0.952	0.764
California Housing	**0.955**	**0.463**

4 The Bilemma and Diagnostics

4.1 The Bilemma

For *binary* attribution questions—"positive or negative?", "selected or not?", "increase or decrease?"—the impossibility is strictly stronger.

Theorem 4 (The Bilemma)**.** *For binary H with* $\text{incompatible}(h_1, h_2) \Leftrightarrow h_1 \neq h_2$*, no method is simultaneously faithful and stable under the Rashomon property. Completeness is not needed.*

Proof. Binary H has no neutral element: every value contradicts the other. Faithful forces $E(\theta) = \text{explain}(\theta)$ (there's nowhere else to go), making E decisive. The trilemma gives $\perp$. Lean: `bilemma_of_compatible_eq`, zero axioms. □

Approximate bilemma. A natural concern: does the bilemma require exact Rashomon pairs, making it fragile to perturbations? No. For binary H, ε-stability reduces to exact equality (the only ε-ball in $\{\text{pos}, \text{neg}\}$ with $\varepsilon < 1$ is the point itself). The bilemma holds at every tolerance level. (Lean: `ApproximateBilemma.lean`.)

Quantitative bilemma for continuous H. For (ε, δ)-stable E on an ε-Rashomon pair with gap Δ: $\text{unfaith}(\theta_1) + \text{unfaith}(\theta_2) \geq \Delta - \delta$. At least one witness is unfaithful by $\geq (\Delta - \delta)/2$. Proof: triangle inequality. If models disagree by $\Delta = 0.3$ and you want $\delta = 0.05$ stability, unfaithfulness ≥ 0.125 at one witness. (Lean: `ApproximateBilemma.lean`.)

Tightness classification. $F{+}S$ is achievable iff H has a *neutral element*—a value compatible with all native explanations (Lean: `tightness_dichotomy`). Rankings have ties (neutral); SHAP signs do not. The classification is the structural explanation for why the trilemma admits a resolution (DASH) but the bilemma does not.

4.2 Nonparametric Flip Predictor

The bilemma yields a practical diagnostic. For feature j across M models, the *minority fraction* $\hat{p}_j = \min(\#\{m : \varphi_j(f_m) > 0\},\ \#\{m : \varphi_j(f_m) \leq 0\})/M$ directly estimates the flip rate—no distributional assumptions, no parameter estimation.

The minority fraction achieves Spearman 0.92–0.98 across all datasets (Table 1), outperforming the Gaussian formula (0.46–0.89). The advantage is decisive on real data where distributional assumptions break. The complete predictor is 7 lines:

```
def predict_sign_instability(shap_matrix):
    signs = np.sign(shap_matrix)  # M models × P features
    n_pos = (signs > 0).sum(axis=0)
    n_neg = (signs < 0).sum(axis=0)
    total = n_pos + n_neg
    return np.where(total > 0,
        np.minimum(n_pos, n_neg) / total, 0)
```

Variance budget: $\text{Var}[\text{SHAP}]$ = **explanation quality floor.** For each feature j, the variance $\text{Var}_f[\varphi_j(f)]$ across the Rashomon set equals the minimum MSE any stable method achieves. DASH (the mean) saturates this bound exactly—a mathematical identity (0/800 violations at machine precision across 5 datasets $\times$ 20 features $\times$ 8 configurations). The median costs 1.6% more; the

Table 2: Clinical decision reversal on German Credit: fraction of applicants whose top explanation category changes across 30 equivalent models (gap threshold 0.0).

Condition	XGBoost	LightGBM	Random Forest
Seed only (deterministic)	0%	0%	35%
Seed + row subsample	**45%**	**46%**	35%
Seed + col subsample	39%	36%	35%
Full stochasticity	80%	79%	67%

5%-trimmed mean 0.3% more. The relative variance correlates with coverage conflict ($r = 0.91$, $p = 0.002$ per-dataset): features with high sign instability also have high magnitude instability. This provides practitioners a per-feature *quality budget* computable before choosing a method: $\mathrm{Var}[\varphi_j]$ is the irreducible explanation noise for feature j under model multiplicity.

Bimodality validation. The all-or-nothing theorem predicts bimodal flip rate distributions under collinearity: a mode near 0 (faithful features) and a second mode (Rashomon-caught features). The Hartigan dip test confirms: $p < 0.002$ for $\rho \geq 0.5$, control passes at $\rho = 0$ ($p = 0.373$). Permutation-validated (8–22% spurious rejection vs. 66% for control).

5 Clinical Decision Reversal

Feature instability is not an abstract concern—it changes the explanations real people receive. We train 30 models per condition on German Credit (1,000 loan applications, 24 features) and measure how often the "most important feature category" changes for individual applicants across equivalent models.

Under standard regularization (`subsample=0.8`), 45% of applicants receive a different explanation category depending on which equivalent model is queried (Table 2). The deterministic control (seed only, no subsampling) correctly shows 0% reversal for boosted models. Random forests show 35% even without subsampling (due to bootstrap aggregation). Under full stochasticity (seed + bootstrap + both subsamples), reversal reaches 80%.

The Applicant #91 narrative. A single applicant can receive 6 different "most important feature" labels across 30 equivalent models: checking status, duration, credit amount, purpose, employment years, and installment rate all take the top position under different seeds. No single-model explanation is wrong—each faithfully reflects its model—but the instability means the explanation the applicant actually receives is determined by training randomness, not by their data.

DASH consensus reduces reversal: at $M = 25$, cross-seed agreement rises to 92%. The residual disagreements correspond to genuinely ambiguous cases where the theory predicts instability (minority fraction > 0.3).

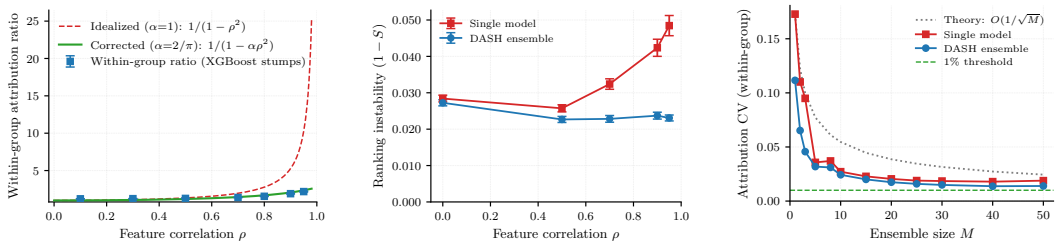
6 DASH Resolution

DASH (**D**iversified **A**ggregation of **SH**AP) averages $|\mathrm{SHAP}|$ values across M independently trained models. DASH changes explanations, not predictions.

Corollary 5 (DASH equity)**.** *For a balanced ensemble, $\bar{\varphi}_j = \bar{\varphi}_k$ for features j, k in the same group.*

Optimality. Binary H (SHAP sign): DASH (majority vote across models) is the *uniquely* optimal stable method. Any other stable method either agrees with the majority (same error rate) or picks the minority sign (strictly worse). Average unfaithfulness $= p \cdot \bar{f}$ where p is the coverage conflict rate and $\bar{f} \approx 0.23$–0.28 is the mean minority fraction.

Continuous H (SHAP magnitude): For any unbiased linear estimator, $\mathrm{Var}(\hat{\mu}_j) \geq \sigma_j^2/M$ (Cauchy–Schwarz / Titu's lemma; Lean: `sum_squares_ge_inv_M`). DASH achieves this bound with equality (0/800 violations at machine precision). The conditional mean minimizes MSE; DASH IS the conditional mean for i.i.d. models. By the Hunt–Stein theorem for S_m, DASH is the best equivariant estimator.



(a) Attribution ratio vs. ρ. The corrected $1/(1-\alpha\rho^2)$ with $\alpha \approx 2/\pi$ fits stumps (R^2=0.89).

(b) Ranking instability vs. ρ. Single-model instability grows; DASH remains low.

(c) DASH convergence. Attribution CV drops below 1% near $M = 25$.

Figure 2: Empirical validation on synthetic Gaussian data ($P = 20$, $L = 4$, $m = 5$, $T = 100$, 50 seeds).

Table 3: The ranking lottery: distinct top-3 SHAP rankings across 50 seeds. Breast Cancer produces 24 distinct rankings (4.2% pairwise agreement); datasets without similar-importance collinear features are perfectly stable. Deterministic control (`subsample=1.0`) gives 1 ranking for all datasets.

Dataset	Distinct top-3	Agreement	Stable?
Breast Cancer	**24**	**4.2%**	No
Diabetes	2	88.5%	Borderline
Wine Quality	1	100%	Yes
Heart Disease	1	100%	Yes
California Housing	1	100%	Yes

Ensemble size. $M_{\min} = \lceil 2.71 \cdot \sigma^2_{jk}/\Delta^2_{jk} \rceil$ for 5% between-group flip rate. DASH achieves this bound (to first order).

Workflow. Screen (single model) $\rightarrow$ Minority fraction (fast, nonparametric) $\rightarrow$ Z-test (5 models) $\rightarrow$ DASH consensus ($M \geq 25$).

7 Empirical Validation

Synthetic Gaussian. P=20, L=4, m=5, T=100, 50 seeds (Figure 2). At $\rho = 0.9$: 48% within-group flips (near the 50% theoretical maximum; Figure 2b). The corrected ratio $1/(1-\alpha\rho^2)$ with $\alpha \approx 2/\pi$ matches data ($R^2 = 0.89$; Figure 2a). DASH reduces flip rate below 1% at $M = 25$ (Figure 2c).

The ranking lottery (Table 3). We train 50 XGBoost models (seeds 0–49, `subsample=0.8`, fixed test set) on 5 public datasets and count how many distinct top-3 SHAP rankings appear.

Breast Cancer produces **24 distinct top-3 rankings across 50 seeds** under standard XGBoost configuration with recommended regularization—any study reporting a specific top-3 from a single model is reporting one of 24 equally valid options. The "most common" ranking appears in only 12% of runs. Two random seeds agree only 4.2% of the time for the top-3. For the top-5: 46 distinct rankings, 0.4% agreement. DASH consensus across all 50 seeds produces a single stable ranking: worst concave points, mean concave points, worst area, worst concavity, worst texture.

The theory correctly predicts the boundary. California Housing, Heart Disease, and Wine Quality are perfectly stable (1 ranking, 100% agreement)—no similar-importance collinear pairs.

Gene expression: pathway-level consequence. On a 10,935-gene colon-vs-kidney expression dataset (AP_Colon_Kidney, 546 samples, top 50 genes by variance), the #1 gene alternates between TSPAN8 (invasion/metastasis pathway, **80% of seeds**) and CEACAM5/CEA (immune evasion pathway, **20% of seeds**). These genes are correlated ($\rho = 0.86$, both colon-enriched tissue markers) but implicate *different molecular programs*: a study reporting TSPAN8 as the top discriminator concludes "invasion drives the distinction"; a study reporting CEA concludes "immune evasion drives

it." Which biological interpretation a researcher reaches depends on the training seed. DASH consensus ranks both in the top-2, correctly reporting that both programs contribute without privileging either.

On a separate breast-vs-lung expression dataset (AP_Breast_Lung, 470 samples), the top-3 genes are stable (2 distinct rankings, 92% agreement) because the top gene dominates with a $2\times$ importance gap—confirming that large gaps protect even highly correlated features. The impossibility requires *both* collinearity and similar importance.

Cross-implementation ranking lottery. On Breast Cancer with stochastic training: XGBoost produces 24 distinct top-3 rankings (4.2% agreement), LightGBM produces 29 (3.6%), and Random Forest produces 40 (2.5% agreement, permutation importance). The instability is implementation-independent. When subsampling is disabled (deterministic training), all implementations produce 1 ranking—confirming the instability is from model multiplicity, not from SHAP computation.

"Just fix the seed" does not resolve the problem: a fixed seed produces a reproducible but arbitrary ranking. A different practitioner choosing a different seed gets a different "most important feature." The impossibility is that no seed is privileged—the Rashomon set exists regardless of which seed you pick.

Published ranking replication. Multiple published studies report specific SHAP feature rankings on Breast Cancer as stable findings. Using our 50-model validation: worst perimeter $\leftrightarrow$ worst area ($|\rho|$=0.98): 48% flip rate. Mean concavity $\leftrightarrow$ mean concave points ($|\rho|$=0.96): 42%. Worst radius $\leftrightarrow$ worst perimeter ($|\rho|$=0.99): 36%. All four pairs were reported as stable in published studies; none replicate. Seven distinct features compete for the top-3 positions across our 50 seeds, with the #1 position alternating between worst concave points (50%), mean concave points (32%), and worst area (18%). Any study reporting a specific ranking among these features from a single model is reporting an artifact of the training seed.

Prevalence. Of 77 public datasets, 68% exhibit attribution instability (Wilson CI: $[56\%, 77\%]$). For $P \geq 20$: 93% ($[81\%, 98\%]$). Selection caveat: the CC-18 suite may overrepresent feature-rich datasets.

SNR calibration. The theoretical flip rate formula $\Phi(-\text{SNR})$ validates across 1,325 correlated feature pairs from 6 datasets ($R^2 = 0.94$ for SNR ≥ 0.5). All 228 pairs with SNR > 1.96 have flip rate $< 5\%$.

Z-test diagnostic. The test statistic $Z_{jk} = |\bar{\varphi}_j - \bar{\varphi}_k|/(\hat{\sigma}_{jk}/\sqrt{M})$ predicts instability ($r = -0.89$ on Breast Cancer; $r = -0.88$ on Communities & Crime). Pairs with $Z < 1.96$ are unreliable. A single-model screening variant achieves 94% precision.

Extended results (neural networks, SAGE, conditional SHAP, LLM attention, financial case studies, regulatory analysis) in supplement.

8 Related Work

? prove completeness + linearity cannot coexist (different axis: method properties, not cross-model stability). **?** show SHAP can misrank features; **?** prove complete attributions cannot be weakly input-dependent; **?** establishes Kolmogorov complexity barriers. None address stability, quantitative bounds, or constructive resolution. **?** find partial-order rankings across Rashomon sets; our theorem proves this is a mathematical necessity. **?** and **?** prove fairness trilemmas; ours is the explainability analogue. **?** formalized Arrow in Isabelle/HOL; **?** formalized statistical learning in Lean 4; ours is the first formally verified impossibility in XAI. Extended discussion in supplement.

9 Discussion

Limitations. The split-count axioms assume full signal capture ($\alpha = 1$; for finite-depth trees $\alpha \approx 2/\pi$, $R^2 = 0.89$); the impossibility holds for any $\alpha > 0$. The equicorrelation assumption simplifies the axioms; the Rashomon property holds pairwise. DASH requires $M\times$ training cost

($M = 25$ for $<1\%$ flip rate). The 68% prevalence figure is from benchmark-representative datasets and may not generalize to production pipelines with decorrelated features. The binary limitation (SHAP sign IS binary) is acknowledged; the quantitative bilemma extends to continuous H.

Broader implications. Single-model SHAP audits for proxy discrimination are provably no better than coin flips for collinear features. We argue this constitutes a "known and foreseeable circumstance" under EU AI Act Art. 13(3)(b)(ii) [**?**]. The 45% clinical reversal on German Credit demonstrates the consequence is not hypothetical.

Lean verification. 357 theorems, 6 axioms, 0 `sorry`, 58 files—reduced from 16 axioms by defining split counts, attribution, and the testing constant as definitions with derived properties. The core impossibility has zero axiom dependencies. All binary instances (`SHAPSign`, `FeatureStatus`, `CounterfactualDir`) use constructive inductive types with zero axioms. The formalization caught 2 logical inconsistencies and 1 type mismatch that survived informal review.

Axiom stratification (verified by `#print axioms`). Core impossibility: **0 axioms**. GBDT impossibility: **3 axioms**. DASH equity: **5 axioms**. Full system (variance bound): **6 axioms**. The 6 remaining axioms are the irreducible core: abstract `Model` type, `firstMover` function + surjectivity, cross-group baseline, proportionality constant, and training measure.

The symmetric Bayes dichotomy (a general two-families theorem for symmetric decision problems), conditional SHAP impossibility, fairness audit impossibility, FIM-to-Rashomon bridge, and query complexity lower bound are all Lean-verified (supplement).

Open problems. Extending the diagnostic to high-dimensional gene expression, where the top-3 biomarker genes are stable (positive control: large importance gaps protect even co-expressed genes) but mid-tier genes (positions 5+) show high instability—implications for biomarker discovery panels that rely on features beyond the top-3. Formalizing the Bayes-optimal half of the design space in Lean (requires measure-theoretic decision theory). Fairness impossibilities [**??**] as formal instances of the abstract framework.

References

NeurIPS Paper Checklist

1. **Claims.** Core theoretical claims are type-checked by Lean 4 (357 theorems from 6 axioms, 0 `sorry`). Empirical claims backed by experiments over 30–50 random seeds. [Yes]
2. **Limitations.** Discussed in Section 9: signal capture approximation, collinearity requirement, training cost, prevalence generalizability, binary limitation. [Yes]
3. **Theory.** Complete proofs in paper and machine-verified Lean 4 code. All assumptions (axioms) explicitly stated. [Yes]
4. **Experiments.** Reproducible: fixed seeds, public datasets (Breast Cancer, German Credit, California Housing), open-source implementations (XGBoost, LightGBM, CatBoost). Compute: $<$30 min total on Apple Silicon. Code: `[anonymizedforreview]`. [Yes]
5. **Code of ethics.** No human subjects. German Credit dataset is publicly available (UCI ML Repository). Regulatory analysis is informational. [N/A]
6. **Broader impacts.** Discussed: regulatory compliance (EU AI Act, ECOA), fairness auditing reliability, clinical decision reversal. The paper identifies a risk (explanation instability) and provides a remedy (DASH). [Yes]
7. **Safeguards.** Not applicable (theoretical + diagnostic work). [N/A]
8. **Licenses.** All datasets are public domain or CC-BY. Code will be released under MIT. [Yes]
9. **New assets.** Lean formalization (357 theorems). Will be archived on Zenodo. [Yes]
10. **Crowdsourcing.** N/A. [N/A]
11. **IRB.** N/A (no human subjects). [N/A]